

QUIZ MASTER V1 DOCUMENTATION

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DESCRIPTION

Project Statement: It is a multi-user app (one requires an administrator and other users) that acts as an exam preparation site for multiple courses.

Quiz Master - V1 is a multi-user exam preparation platform where an admin manages subjects and quizzes, and students can register, enrol in subjects, attempt quizzes, and view their results. The platform ensures a smooth user experience with features such as score tracking, downloadable transcripts, and a well-organized dashboard.

This project is designed with scalability and user engagement in mind, and intuitive navigation to enhance the learning experience.

TECHNOLOGIES USED

Backend

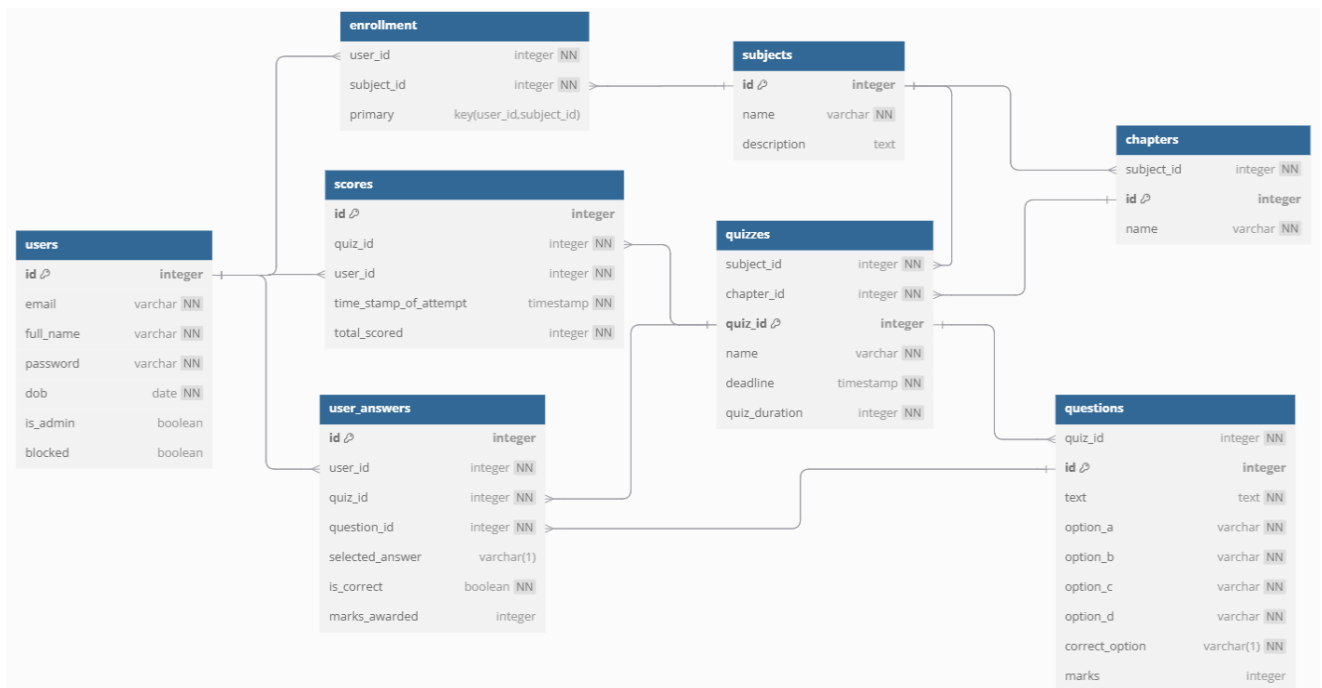
- Flask (Web framework)
- Flask-Login (User authentication)
- Flask-SQLAlchemy (ORM for database management)
- SQLite (Database)
- Flask-WTF (Form handling)
- Flask-Migrate (Database migrations)

Frontend

- Jinja2 (Templates)
 - HTML, CSS, Bootstrap (UI design)
 - Matplotlib & Seaborn (For generating reports and visualizations)
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DATABASE DESIGN

- **Relational Structure:** Includes well-defined tables for users, subjects, quizzes, questions, attempts, and scores, linked via foreign keys.
- **Role-based Access Control:** Differentiates between admin and student access, ensuring secure and structured data operations.
- **User Management:** Stores admin and student details, including authentication and enrolment information.
- **Quiz & Attempt Tracking:** Records quiz details, deadlines, student attempts, scores, and timestamps for performance monitoring.
- **Data Integrity & Consistency:** Maintains structured data relationships with cascade deletion to ensure consistency and enable seamless retrieval for analytics and transcripts.



ARCHITECTURE AND FEATURES

The project follows a **Model-View-Controller (MVC) architecture** for better modularity and maintainability.

- **Controller:** The app.py file contains the core application logic, including routing, user authentication, and request handling. The file extensions.py contains the initialization for the same.
- **Model:** The database models are defined in models.py using Flask-SQLAlchemy, to store the users, subjects, chapters, quizzes and the respective questions and scores, and keep track of enrolments.
- **View:** The user interface is managed through Jinja2 templates located in the templates/ directory. These templates dynamically render content for different user roles (admin and student).
- **Static Files:** CSS files and images are stored in the static/ directory.

FEATURES IMPLEMENTED

The platform enables admins to efficiently create and manage subjects, quizzes, and student enrolments with role-based access control to ensure a clear distinction between admin and student functionalities. A structured quiz system ensures that students can attempt quizzes within a set deadline and duration, with automatic submission upon expiration. Scores are calculated instantly, stored in the database, and accessible to both students and admins. Students can track their performance with detailed quiz history and progress insights. Admins can monitor student activity, generate PDF transcripts, and enforce quiz deadlines to maintain fairness.

Additionally, the platform includes interactive analytics for admins to visualize student performance trends. An enrolment system has also been implemented that restricts students to only their assigned subjects, preventing unauthorized access. The interface is optimized for usability, with smooth navigation, animations, and a high-contrast design for an enhanced user experience.

VIDEO DEMONSTRATION

<https://drive.google.com/file/d/1xDv5VZ1RA3fGUiRckXXHWDDRxdUcYNql/view?usp=sharing>